Meta Heuristic Optimization

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Objective Function

- Optimization problems can be approximated to a mathematical function called objective function
- Ex.

$$\min f(\mathbf{x})$$

and this may be subject to constraints

(hard constraints) $g_i(\mathbf{x}) = c_i, \quad i = 1, \ldots, n$

(inequality constraints) $h_j(\mathbf{x}) \geq d_j, \quad j = 1, \ldots, m$

Mathematical formulation

If we know $f'(\mathbf{x})$ (the first derivative of $f(\mathbf{x})$) we could find the minima of the objective function by finding all $\mathbf{x} = \mathbf{x}^*$ such that

$$f'(\mathbf{x}^*) = 0$$

We can verify that this is a minima or maxima by considering the sign of the second derivative. I.e.,

$$f''(\mathbf{x}^*) < 0 \qquad \text{maxima at } \mathbf{x}^*$$
$$f''(\mathbf{x}^*) > 0 \qquad \text{minima at } \mathbf{x}^*$$

Mathematical formulation

Unfortunately, for many practical problems, this is intractable… or we cannot compute $f'(\mathbf{x})$

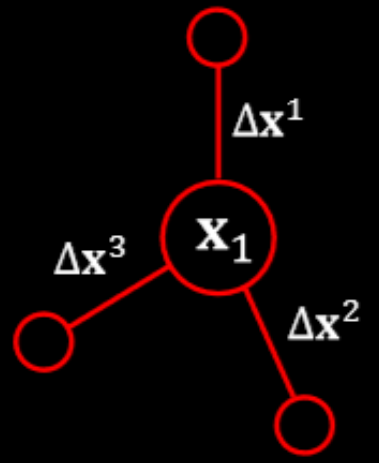
- An alternative strategy is then to generate a set of *feasible solutions* $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ and to evaluate each of these, producing
$$\{f(\mathbf{x}_1), f(\mathbf{x}_2), \ldots, f(\mathbf{x}_n)\}$$

- We can then find $\min\{f(\mathbf{x}_1), f(\mathbf{x}_2), \ldots, f(\mathbf{x}_n)\}$
and hope that this is the true solution!

What could possibly go wrong!?

Gradient descent (Hill climbing)

For example, consider an initial feasible solution $\mathbf{x}_1$



We then consider small changes in this solution…

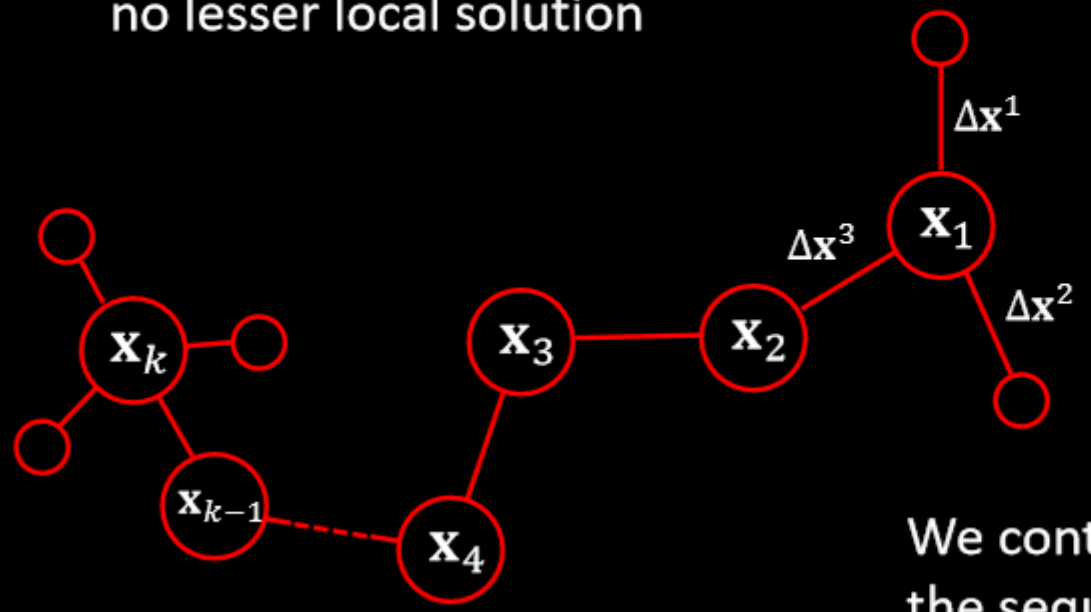
…and find a better feasible solution as

$$\mathbf{x}_2 = \min\{\mathbf{x}_1 + \Delta\mathbf{x}^1, \mathbf{x}_1 + \Delta\mathbf{x}^2, \mathbf{x}_1 + \Delta\mathbf{x}^3\}$$

Gradient descent (hill climbing)

- For example, consider an initial feasible solution $\mathbf{x}_1$

We terminate this search when we find no lesser local solution



We then consider small changes in this solution…

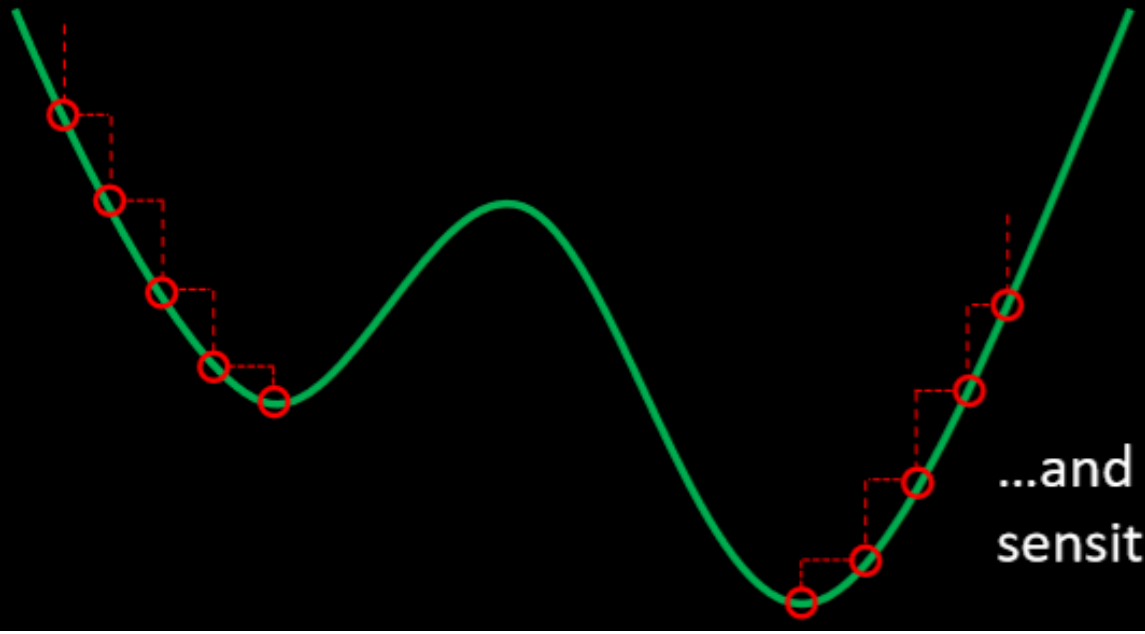
…and find a better feasible solution as

$$\mathbf{x}_2 = \min\{\mathbf{x}_1 + \Delta\mathbf{x}^1, \mathbf{x}_1 + \Delta\mathbf{x}^2, \mathbf{x}_1 + \Delta\mathbf{x}^3\}$$

We continue this procedure producing the sequence $\mathbf{x}_1 > \mathbf{x}_2 > \cdots > \mathbf{x}_k$

Gradient descent (hill climbing)

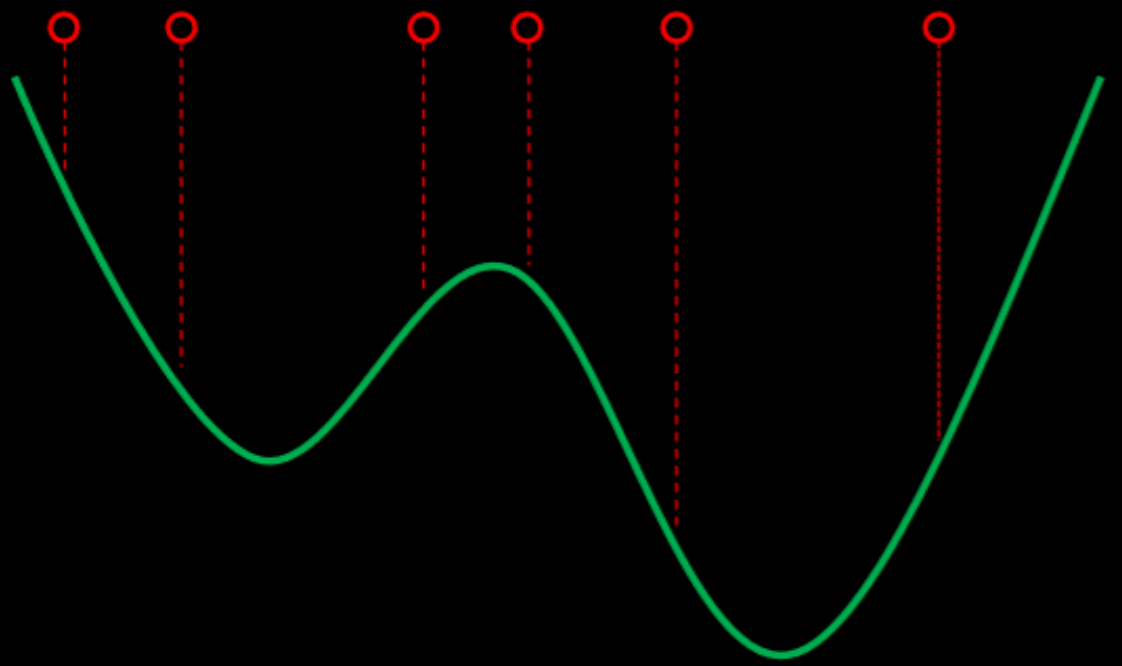
- Gradient Descent is an iterative method… it generates a sequence of minimisers, with the most recent being the best solution found so far
- Unfortunately, it doesn't work for many problems…



…and the results may be sensitive to initial conditions

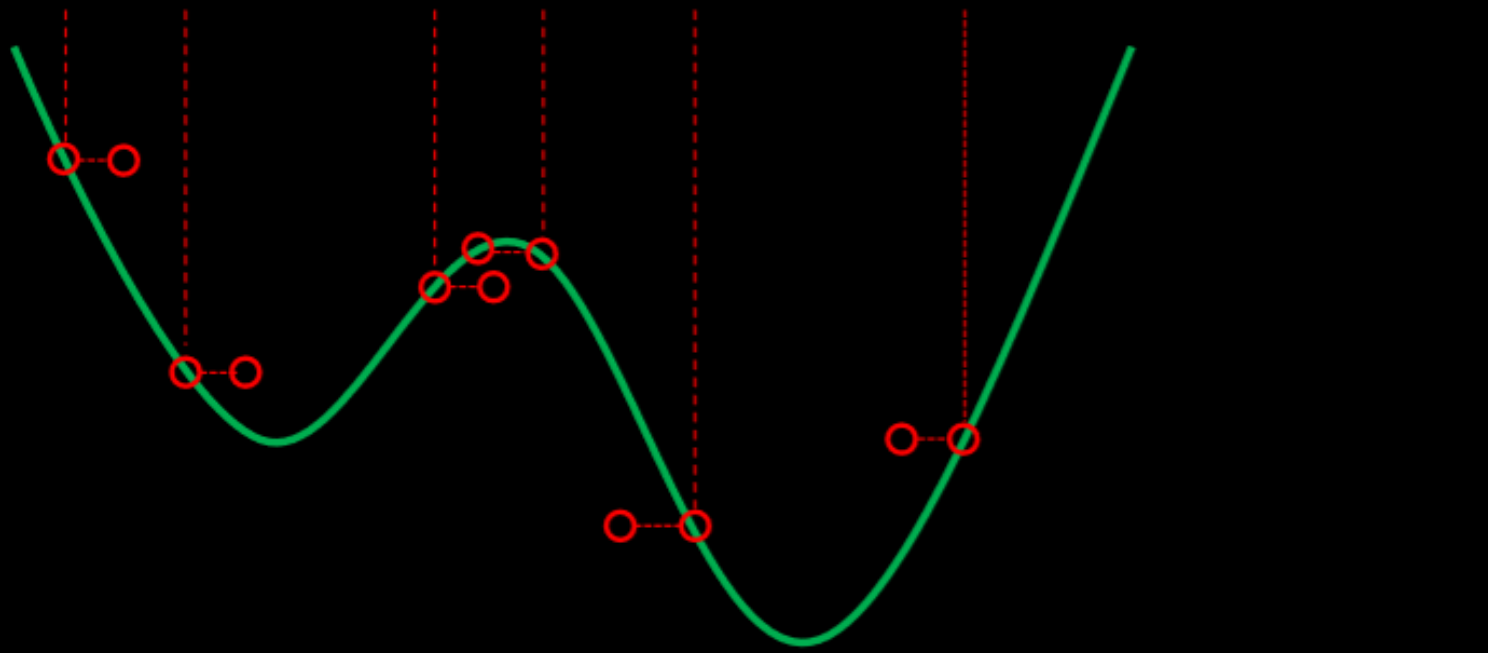
Meta heuristic optimization

- In such situations, another strategy is to start with a *population* of feasible solutions



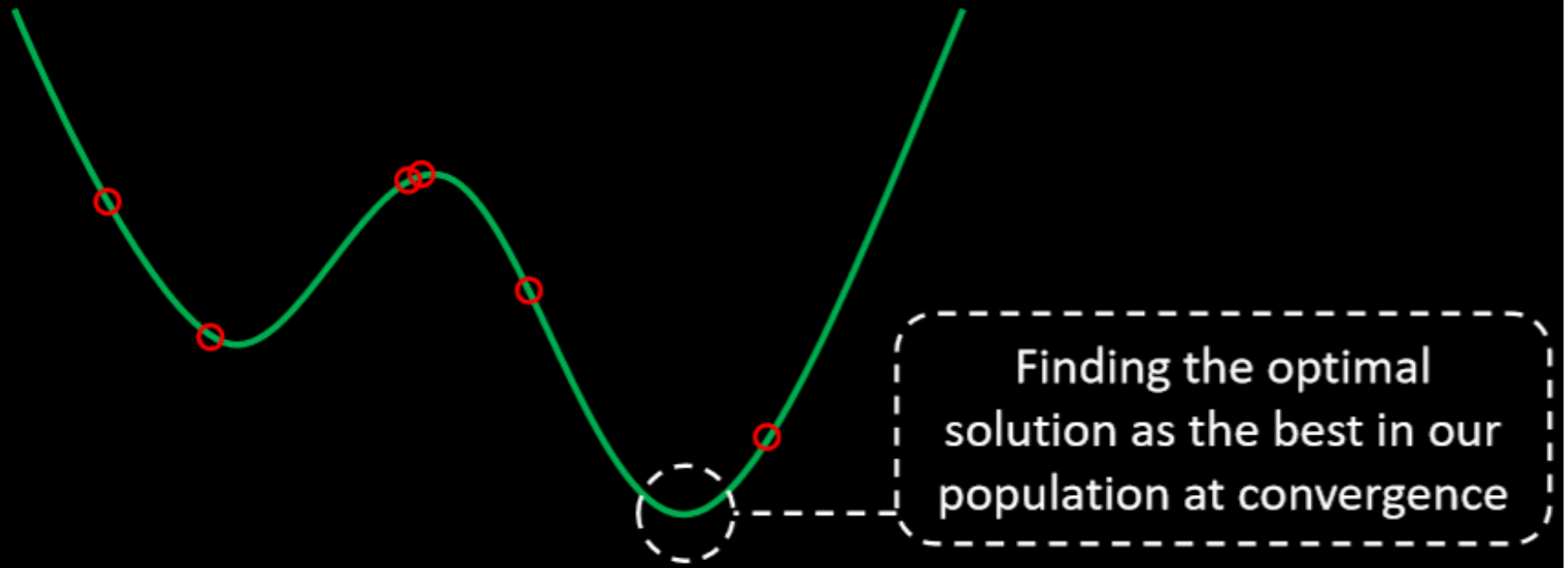
Meta heuristic optimization

- We then generate new solutions in the neighbourhood of the previous candidates…



Meta heuristic optimization

- …and continue this procedure until we cannot find further improvements!

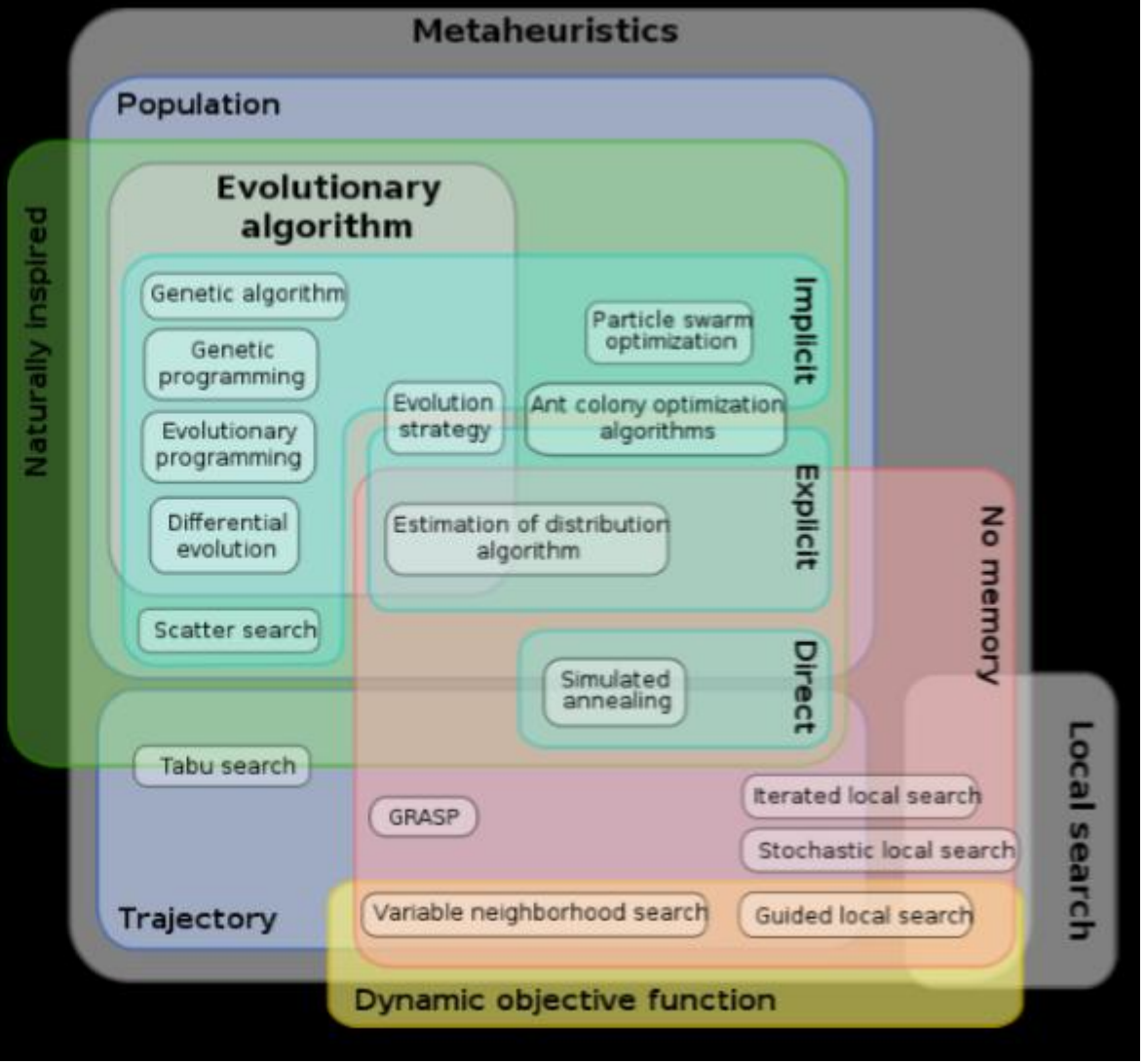


Scope of meta-heuristics

- There are a large number of optimisation algorithms based on meta-heuristics…

  …we will present one herein, based on evolutionary principles…

  …the *genetic algorithm*



Genetic Algorithms

- A feasible solution is encoded in a finite string of characters taken from some alphabet

$$\mathbf{x} = \{x_1, x_2, \ldots, x_m\}$$

$$\overbrace{aacbca}\cdots\overbrace{bcabba}$$

- The objective function of the optimisation problem is called a *fitness function* for an evolutionary algorithm

$$f(\mathbf{x}) = f(aacbca \cdots bcabba)$$

Genetic Algorithms

- Each string is a candidate solution known as a *phenotype*

Each element of the string is called a *gene*

The characters for a gene are called *allele* values of the gene

The sequence of genes is called a *chromosome*

$aacbcabcabba$	$f(\mathbf{x}_1)$
$babbaaababcc$	$f(\mathbf{x}_2)$
$bcbaababaacb$	$f(\mathbf{x}_3)$
…	…
$caaccbccbcaa$	$f(\mathbf{x}_m)$

Each phenotype has a *fitness value*

Evolution of the population is controlled by *genetic operators*

Holland's Genetic Algorithm

- In Holland's algorithm, the chromosomes are binary strings

  10111010101
  11001110000
  01010100111
  $\ldots$
  10111001100

- The genetic operators are: *selection*, *crossover* and *mutation*

Selection

- Suppose the population contains the following strings with the given fitness values:

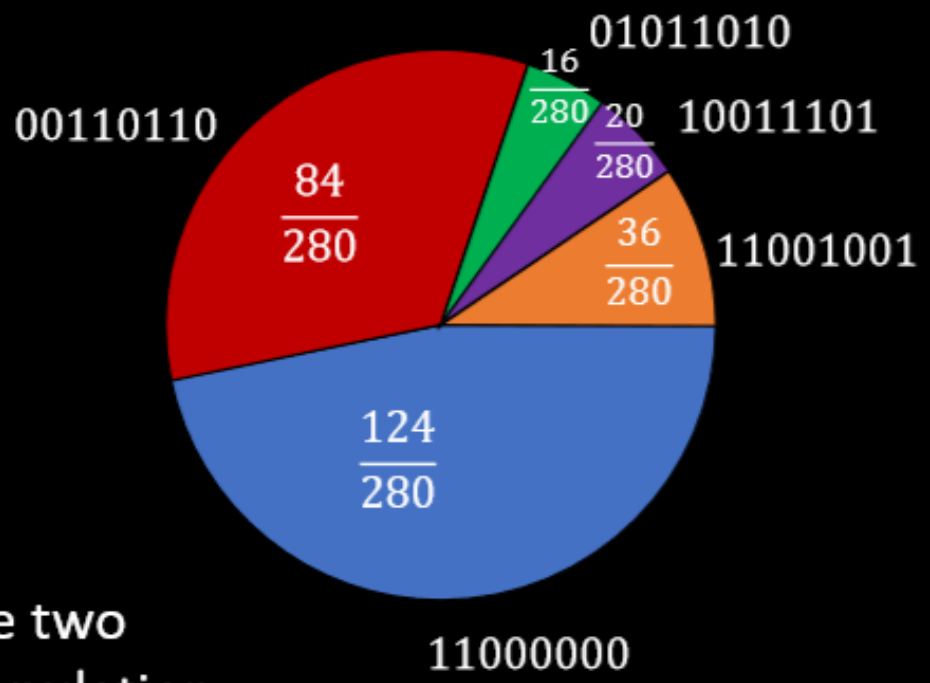
10011101	20
11001001	36
01011010	16
00110110	84
11000000	124

- Selection involves choosing two strings for breeding (reproduction)
  - This is done with a probability proportionate to the relative fitness of the strings… known as roulette wheel selection

Selection

- In roulette wheel selection, each string receives a number of slots on a wheel equal to their fitness

10011101	20
11001001	36
01011010	16
00110110	84
11000000	124
total:	280



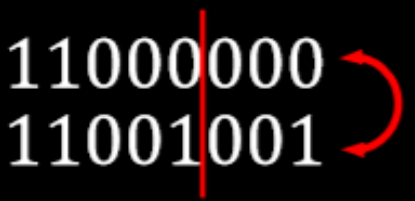
The wheel is spun to generate two (unique) samples from the population

Crossover

- Given two samples taken from the population by selection…

11000000
11001001

we select randomly (with uniform probability) a site between genes



we then swap the tails of these two strings, to produce offspring

11000001
11001000

Mutation

- Finally we test each gene in each offspring phenotype for mutation
- With very small probability p_m we randomly select a new allele value for that gene
- E.g.

$$11000001$$
$$11001000$$

- We then compute the fitness value of these two offspring, and insert them into the population

Holland's Genetic Algorithm

- Finally, we eliminate (kill off) the weakest individuals in the population, keeping a constant population size

~~10011101~~	~~20~~
11001001	36
~~01011010~~	~~16~~
00110110	84
11000000	124
11010001	112
11001000	96

Termination of a GA

- The generational process is repeated until a termination condition has been reached.
- Common terminating conditions are:
  - A solution is found that satisfies minimum criteria
  - Fixed number of generations reached
  - Allocated budget (computation time/money) reached
  - The highest ranking solution's fitness is reaching or has reached a plateau such that successive iterations no longer produce better results
  - Manual inspection
  - Any Combinations of the above

Exercise 01

- Suppose you need to find the maximum x^2 given $0 <= x < 32$.
  - Write the objective function
  - Generate four phenotypes
  - Complete one iteration by hand

Exercise 01

- $f(x) = \{MAX(x^2): 0 < x < 32 \}$
- Encode Solution: Just use 5 bits (1 or 0).
- Generate initial population.

A	0	1	1	0	1
B	1	1	0	0	0
C	0	1	0	0	0
D	1	0	0	1	1

- Evaluate each solution against objective.

Sol.	String	Fitness	% of Total
A	01101	169	14.4
B	11000	576	49.2
C	01000	64	5.5
D	10011	361	30.9

Exercise 01 contd’

- Create next generation of solutions
  - Probability of “being a parent” depends on the fitness.
- Ways for parents to create next generation
  - Reproduction
    - Use a string again unmodified.
  - Crossover
    - Cut and paste portions of one string to another.
  - Mutation
    - Randomly flip a bit.
  - COMBINATION of all of the above.

Exercise 02

- Consider the Traveling Salesman Problem with 7 cities
  - Formulate a chromosome
  - Hand-execute one iteration for 10 number of chromosomes
- What problems did you encounter?
- How do you think you can solve them?

Random Keys Algorithm

- Consider a TSP example with 5 cities
  - How to represent a solution?
    - (4 2 1 5 3)
    - Path: 3 -> 2 -> 5 -> 1 -> 4
  - Using one-point crossover is difficult
    - Ex:
      - Parents: (1 4 3 2 5), (2 4 5 3 1)
      - Children: (1 4 3 3 1), (2 4 5 1 4)
    - Random Key method: An alternative

Random key method

- Assign a random number drawn uniformly from (0, 1)
  - Decode in ascending order (could use descending as well)
  - Example:

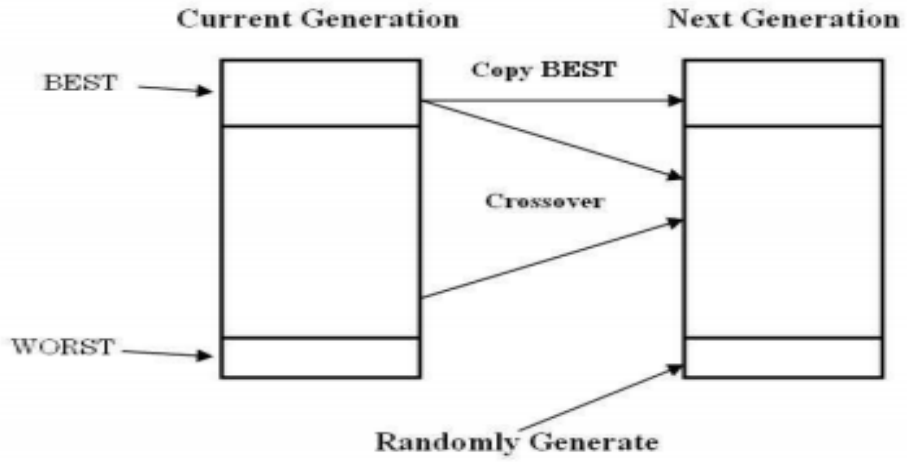
Random key	0.42	0.06	0.38	0.48	0.81
Decodes as	3	1	2	4	5

    - Path: 2 -> 3 -> 1 -> 4 -> 5
  - Crossover operator:
    - Different methods could be used
    - Ex. Parameterized uniform crossover:
      - Select two parents randomly inheriting each gene from parent 1 with probability p and from parent 2 with probability 1 – p

Parent 1	0.56	0.81	0.22	0.7	0.86
Parent 2	0.29	0.49	0.98	0.12	0.32
Offspring	0.59	0.49	0.22	0.12	0.86

Random key method

- Mutation operator
  - Replace the worst set with a newly created set of candidates
    - Randomly generate a new set of individuals – random key method is used
    - Avoids pre-mature convergence
    - Also called immigration



Exercise 01

1. Suppose there is a set of blocks, each with different weights (w) and values (v). There is a backpack that can hold only up to a specific limit of weight (W). You are supposed to fill the backpack with those blocks as it maximizes your value without exceeding the weight limit of the bag. Assuming you are going to use a simple genetic algorithm to solve this problem:

    a. Identify a suitable structure for the chromosome

    b. State the fitness function

    c. Complete the first iteration of the genetic algorithm for a population of four chromosomes. Assume the bag capacity is 15kg and you have 1, 3, 5, 7, 9 and 12 kg blocks with 100, 300, 200, 500, 150 and 400 Rupee values. You are required to clearly demonstrate the selection, crossover and mutation processes

Exercise 02

1. ***'The Spinning Wheel'*** is a large and popular amusement park that has several attractions. The management has done a survey and identified the average satisfaction level the visitors can have at each attraction. However, due to excessive demand, each attraction has a waiting time which can range from few minutes to couple of hours. The amusement park is open from 8.00 AM to 6.00 PM but the visitors can enter only till 3.00 PM. The management wishes to provide an optimal plan for visitors to visits the attractions optimizing their satisfaction while managing the remaining time for them till the closure of the park. Do you think a genetic algorithm based solution could be used for this task? Justify your answer.

Multi-objective Optimization

- Involves more than one objective function
- Ex:

$$\textbf{Minimize :}$$
$$f_1(x) = x_1^2 - x_1x_2 + x_2^2 - x_1 - x_2 + 1$$
$$f_2(x) = x_1^2 + x_1x_2 + x_2^2 + 2x_1 + 4x_2 + 4$$

- Is it possible to solve this using GA? Give it a try.